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JEE MAIN · QUESTION PAPER

JEE Main 2024 (Online) 27th January Evening Shift

JEE Main Question Paper with Answer Key

27 Jan 2024	90	English
HELD ON	QUESTIONS	LANGUAGE

INSIDE THIS PDF

Chemistry	30 questions
Mathematics	30 questions
Physics	30 questions
Answer key	at the end

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Chemistry 30 questions

Q1. MCQ (Single Correct Answer)

Identify the incorrect pair from the following :

(A) Haber process - Iron

(B) Polythene preparation - $TiCl_4, Al(CH_3)_3$

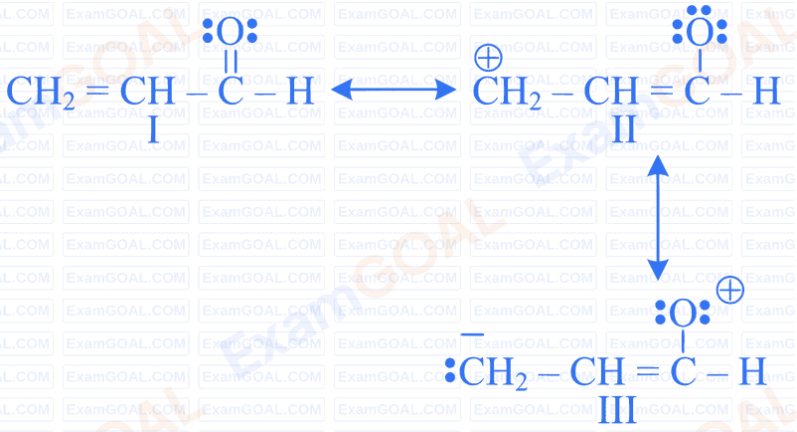
(C) Photography - AgBr

(D) Wacker process - $PtCl_2$

Open solution

Q2. MCQ (Single Correct Answer)

The order of relative stability of the contributing structure is :



Choose the correct answer from the options given below :

(A) I = II = III (B) I > II > III (C) III > II > I (D) II > I > III

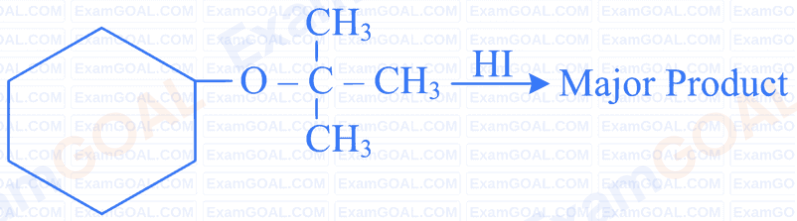
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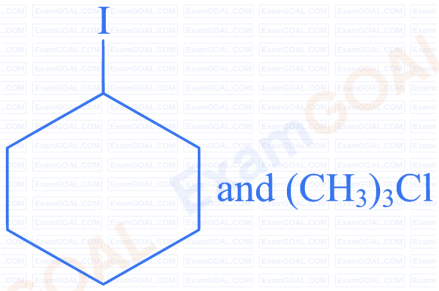
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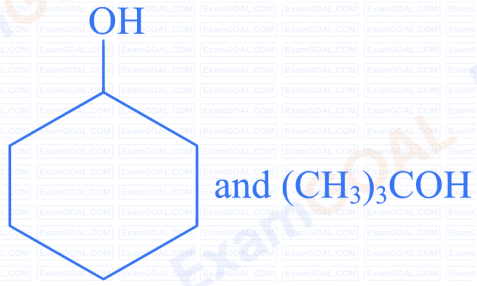
Q3. MCQ (Single Correct Answer)

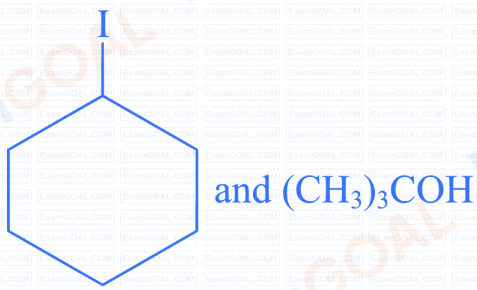
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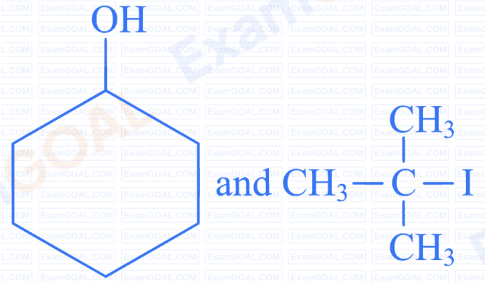
Major product formed in the following reaction is a mixture of :



(A) 

(B) 

(C) 

(D) 

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Q4. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I) : Oxygen being the first member of group 16 exhibits only -2 oxidation state.

Statement (II) : Down the group 16 stability of +4 oxidation state decreases and +6 oxidation state increases.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Both Statement I and Statement II are incorrect

(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are correct

Q5. MCQ (Single Correct Answer)

Open solution

The technique used for purification of steam volatile water immiscible substances is :

(A) Fractional distillation

(B) Distillation

(C) Fractional distillation under reduced pressure

(D) Steam distillation

Q6. MCQ (Single Correct Answer)

Open solution

Which structure of protein remains intact after coagulation of egg white on boiling?

(A) Quaternary (B) Primary (C) Secondary (D) Tertiary

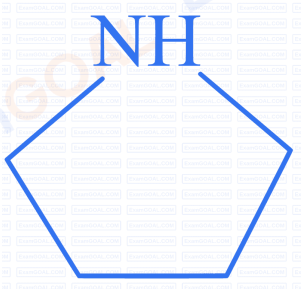
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Q7. MCQ (Single Correct Answer)

Identify B formed in the reaction.

$$Cl-(CH_2)_4-Cl \xrightarrow{\text{excess } NH_3} A \xrightarrow{NaOH} B + H_2O + NaCl$$

Open solution

(A) 

(B) 

(C) $\overset{-}{Cl}\overset{+}{N}H_3-(CH_2)_4-\overset{+}{N}H_3Cl^-$

(D) $H_2N-(CH_2)_4-NH_2$

Q8. MCQ (Single Correct Answer)

Identify from the following species in which d^2sp^3 hybridization is shown by central atom :

Open solution

(A) $[Co(NH_3)_6]^{3+}$

(B) SF_6

(C) $[Pt(Cl_4)]^{2-}$

(D) BrF_5

Q9. MCQ (Single Correct Answer)

Phenolic group can be identified by a positive:

Open solution

(A) Tollen's test

(B) Phthalein dye test

(C) Carbylamine test

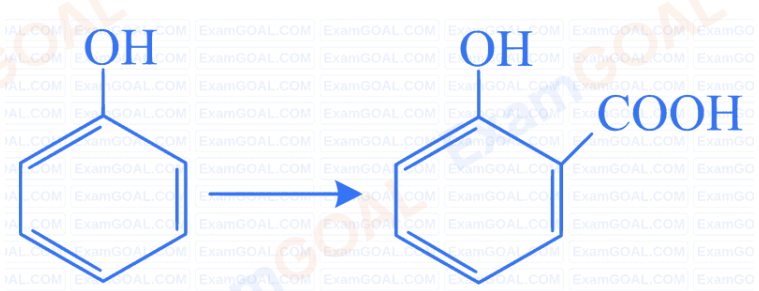
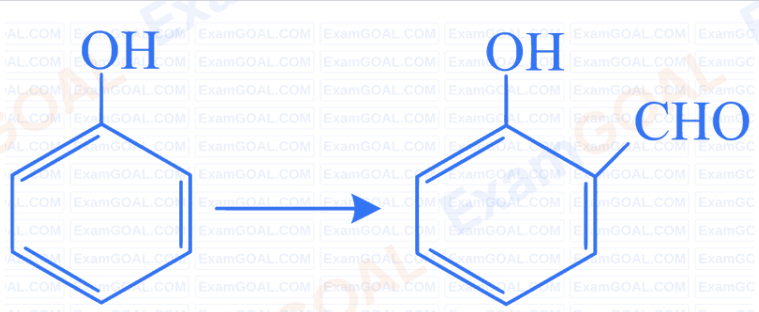
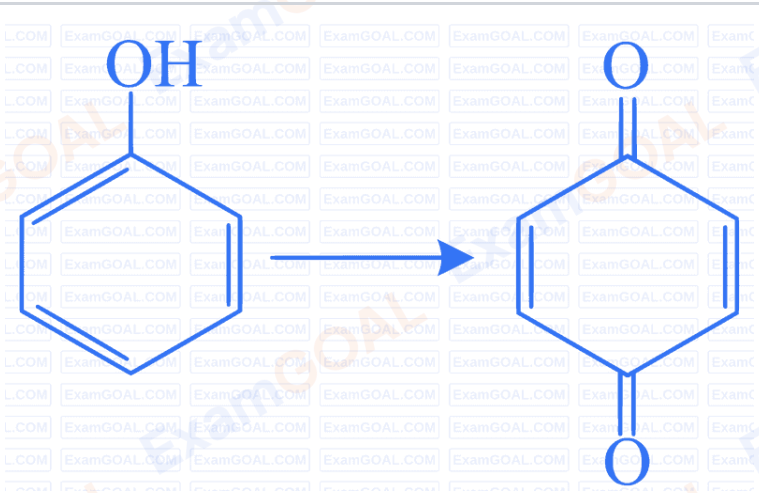
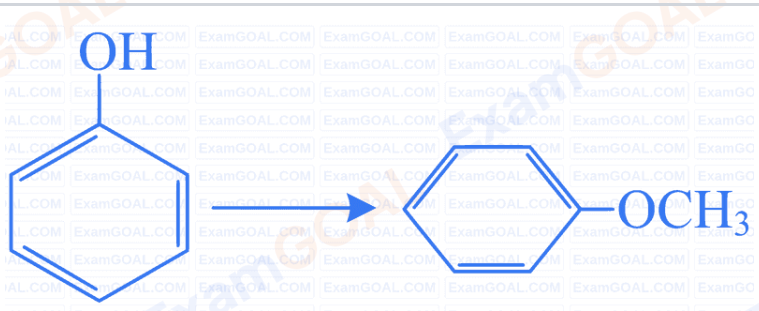
(D) Lucas test

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Q10. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

	List - I (Reaction)		List - II (Reagent(s))
(A)		(I)	$Na_2Cr_2O_7, H_2SO_4$
(B)		(II)	(i) NaOH, (ii) CH_3Cl
(C)		(III)	(i) NaOH, $CHCl_3$, (ii) NaOH, (iii) HCl
(D)		(IV)	(i) NaOH, (ii) CO_2, (iii) HCl

Choose the correct answer from the options given below :

(A) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

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(B) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(C) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(D) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Q11. MCQ (Single Correct Answer)

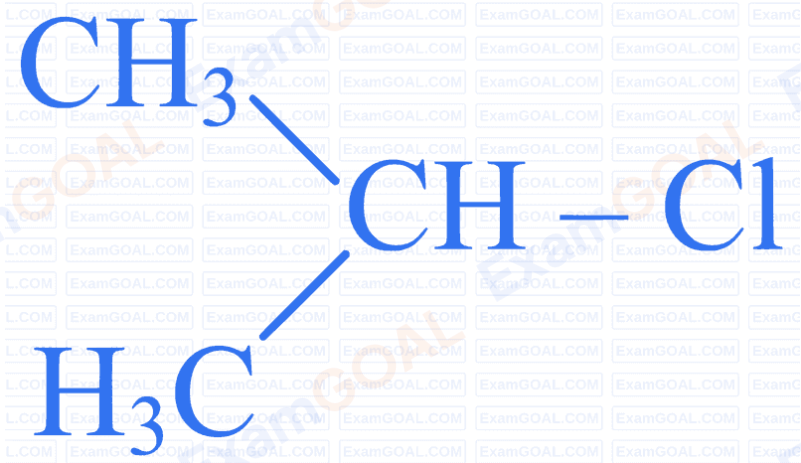
Open solution

Which among the following halide/s will not show S_N1 reaction:

(A) $H_2C = CH - CH_2Cl$

(B) $CH_3 - CH = CH - Cl$

(C) 

(D) 

Choose the most appropriate answer from the options given below :

(A) (B) and (C) only

(B) (A), (B) and (D) only

(C) (B) only

(D) (A) and (B) only

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Q12. MCQ (Single Correct Answer)

Which of the following statements is not correct about rusting of iron?

(A) Rusting of iron is envisaged as setting up of electrochemical cell on the surface of iron object.

(B) Dissolved acidic oxides SO_2, NO_2 in water act as catalyst in the process of rusting.

(C) Coating of iron surface by tin prevents rusting, even if the tin coating is peeling off.

(D) When pH lies above 9 or 10, rusting of iron does not take place.

Open solution

Q13. MCQ (Single Correct Answer)

Which of the following cannot function as an oxidising agent?

(A) SO_4^{2-} (B) MnO_4^- (C) N^{3-} (D) BrO_3^-

Open solution

Q14. MCQ (Single Correct Answer)

Given below are two statements :

Statement (I) : In the Lanthanoids, the formation of Ce^{+4} is favoured by its noble gas configuration.

Statement (II) : Ce^{+4} is a strong oxidant reverting to the common +3 state.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Statement I is false but Statement II is true

Open solution

Q15. MCQ (Single Correct Answer)

Choose the correct option having all the elements with d^{10} electronic configuration from the following :

(A) $^{46}Pd, ^{28}Ni, ^{26}Fe, ^{24}Cr$

(B) $^{29}Cu, ^{30}Zn, ^{48}Cd, ^{47}Ag$

(C) $^{27}Co, ^{28}Ni, ^{26}Fe, ^{24}Cr$

(D) $^{28}Ni, ^{24}Cr, ^{26}Fe, ^{29}Cu$

Open solution

Q16. MCQ (Single Correct Answer)

The incorrect statement regarding conformations of ethane is :

(A) The dihedral angle in staggered conformation is 60°.

(B) Ethane has infinite number of conformations.

(C) Eclipsed conformation is the most stable conformation.

(D) The conformations of ethane are inter-convertible to one-another.

Open solution

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Q17. MCQ (Single Correct Answer)

Open solution

The final product A, formed in the following reaction sequence is:

$$Ph-CH=CH_2 \xrightarrow[\text{(iii) HBr}]{\text{(i) } BH_3,\ \text{(ii) } H_2O_2,\ \Theta OH} A \quad \text{(iv) Mg, ether, then } HCHO/H_3O^+$$

(A) $Ph-\underset{\displaystyle CH_2OH}{\underset{|}{CH}}-CH_3$

(B) $Ph-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

(C) $Ph-CH_2-CH_2-CH_2-OH$

(D) $Ph-CH_2-CH_2-CH_3$

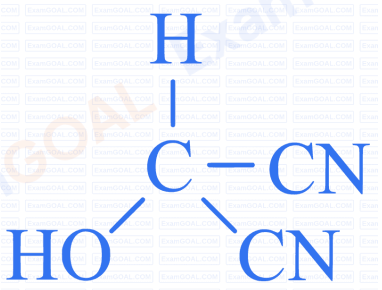
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Q18. MCQ (Single Correct Answer)

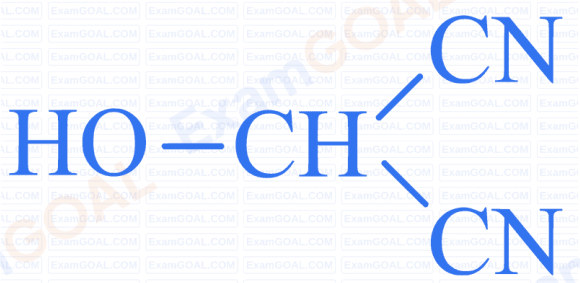
Bond line formula of $HOCH(CN)_2$ is :

Open solution

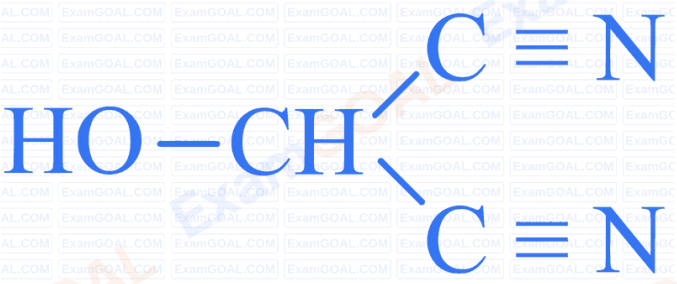
(A)



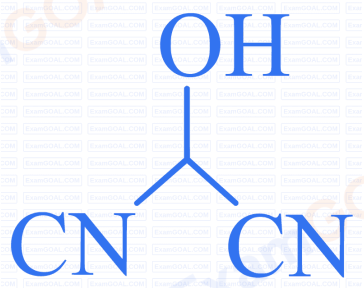
(B)



(C)



(D)



Q19. MCQ (Single Correct Answer)

The quantity which changes with temperature is :

(A) Molality (B) Molarity (C) Mole fraction (D) Mass percentage

Open solution

Q20. MCQ (Single Correct Answer)

The molecular formula of second homologue in the homologous series of mono carboxylic acids is

(A) $C_2H_4O_2$ (B) $C_2H_2O_2$

(C) CH_2O (D) $C_3H_6O_2$

Open solution

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Q21. Numerical

1 mole of PbS is oxidised by "X" moles of O_3 to get "Y" moles of O_2. $X + Y =$ _____.

Open solution

Q22. Numerical

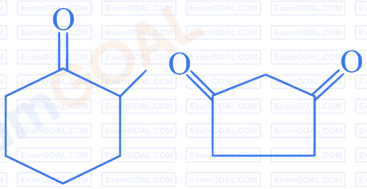
The Spin only magnetic moment value of square planar complex $[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$ is _____ B.M. (Nearest integer)

(Given atomic number for $Pt = 78$)

Open solution

Q23. Numerical

Total number of compounds with Chiral carbon atoms from following is _____.



$CH_3-CH_2-CH(NO_2)-COOH$

$CH_3-CH_2-CHBr-CH_2-CH_3$

$CH_3-CH(I)-CH_2-NO_2$

$CH_3-CH_2-CH(OH)-CH_2OH$

$CH_3-\underset{\underset{I}{|}}{CH}-CH(I)-C_2H_5$

Open solution

Q24. Numerical

Time required for completion of 99.9% of a First order reaction is _____ times of half life $(t_{1/2})$ of the reaction.

Open solution

Q25. Numerical

The hydrogen electrode is dipped in a solution of $pH = 3$ at $25^\circ C$. The potential of the electrode will be _____ $\times 10^{-2}$ V.

$\left(\frac{2.303RT}{F} = 0.059\ V\right)$

Open solution

Q26. Numerical

The number of non-polar molecules from the following is _____.

$HF, H_2O, SO_2, H_2, CO_2, CH_4, NH_3, HCl, CHCl_3, BF_3$

Open solution

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Q27. Numerical

$9.3\ \mathrm{g}$ of aniline is subjected to reaction with excess of acetic anhydride to prepare acetanilide. The mass of acetanilide produced if the reaction is 100% completed is _____ $\times 10^{-1}\ \mathrm{g}$.

(Given molar mass in $\mathrm{g\ mol^{-1}}$

$\mathrm{N}: 14, \mathrm{O}: 16,$
$\mathrm{C}: 12, \mathrm{H}: 1$)

Open solution

Q28. Numerical

Volume of $3\mathrm{M\ NaOH}$ (formula weight $40\ \mathrm{g\ mol^{-1}}$) which can be prepared from $84\ \mathrm{g}$ of NaOH is _____ $\times 10^{-1}\mathrm{dm}^3$.

Open solution

Q29. Numerical

Total number of ions from the following with noble gas configuration is _____.

$\mathrm{Sr}^{2+}(z=38), \mathrm{Cs}^{+}(z=55), \mathrm{La}^{2+}(z=57), \mathrm{Pb}^{2+}(z=82), \mathrm{Yb}^{2+}(z=70)$ and $\mathrm{Fe}^{2+}(z=26)$

Open solution

Q30. Numerical

For a certain thermochemical reaction $\mathrm{M} \rightarrow \mathrm{N}$ at
$\mathrm{T}=400\ \mathrm{K}, \Delta \mathrm{H}^{\ominus}=77.2\ \mathrm{kJ\ mol^{-1}}, \Delta \mathrm{S}=122\ \mathrm{JK^{-1}}$, log equilibrium constant $(\log K)$ is _____ $\times 10^{-1}$.

Open solution

Mathematics 30 questions

Q31. MCQ (Single Correct Answer)

Considering only the principal values of inverse trigonometric functions, the number of positive real values of x satisfying $\tan^{-1}(x)+\tan^{-1}(2x)=\frac{\pi}{4}$ is :

(A) more than 2 (B) 2 (C) 0 (D) 1

Open solution

Q32. MCQ (Single Correct Answer)

Let the position vectors of the vertices A, B and C of a triangle be $2\hat{i}+2\hat{j}+\hat{k}, \hat{i}+2\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}+2\hat{k}$ respectively. Let l_1, l_2 and l_3 be the lengths of perpendiculars drawn from the orthocentre of the triangle on the sides AB, BC and CA respectively, then $l_1^2+l_2^2+l_3^2$ equals:

(A) $\frac{1}{4}$ (B) $\frac{1}{5}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

Open solution

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Q33. MCQ (Single Correct Answer)

Consider the function $f:(0,2) \to \mathbf{R}$ defined by $f(x)=\frac{x}{2}+\frac{2}{x}$ and the function $g(x)$ defined by $g(x)=\begin{cases}\min\{f(t)\}, & 0<t\leq x \text{ and } 0<x\leq 1 \\ \frac{3}{2}+x, & 1<x<2\end{cases}$. Then,

Open solution

(A) g is continuous but not differentiable at $x=1$

(B) g is continuous and differentiable for all $x \in (0,2)$

(C) g is not continuous for all $x \in (0,2)$

(D) g is neither continuous nor differentiable at $x=1$

Q34. MCQ (Single Correct Answer)

An urn contains 6 white and 9 black balls. Two successive draws of 4 balls are made without replacement. The probability, that the first draw gives all white balls and the second draw gives all black balls, is :

Open solution

(A) $\frac{3}{256}$ (B) $\frac{5}{256}$ (C) $\frac{3}{715}$ (D) $\frac{5}{715}$

Q35. MCQ (Single Correct Answer)

Let the image of the point $(1,0,7)$ in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ be the point (α,β,γ). Then which one of the following points lies on the line passing through (α,β,γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with y-axis and z-axis respectively and an acute angle with x-axis ?

Open solution

(A) $(1,-2,1+\sqrt{2})$ (B) $(3,-4,3+2\sqrt{2})$ (C) $(3,4,3-2\sqrt{2})$ (D) $(1,2,1-\sqrt{2})$

Q36. MCQ (Single Correct Answer)

Let A and B be two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B. Then the distance of the point $P(m,n)$ from the point $Q(-2,-3)$ is :

Open solution

(A) 8 (B) 10 (C) 4 (D) 6

Q37. MCQ (Single Correct Answer)

If α,β are the roots of the equation, $x^2-x-1=0$ and $S_n=2023\alpha^n+2024\beta^n$, then :

Open solution

(A) $2S_{12}=S_{11}+S_{10}$ (B) $S_{12}=S_{11}+S_{10}$ (C) $S_{11}=S_{10}+S_{12}$ (D) $2S_{11}=S_{12}+S_{10}$

Q38. MCQ (Single Correct Answer)

Let e_1 be the eccentricity of the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ and e_2 be the eccentricity of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b$, which passes through the foci of the hyperbola. If $e_1e_2=1$, then the length of the chord of the ellipse parallel to the x-axis and passing through $(0,2)$ is :

Open solution

(A) $\frac{8\sqrt{5}}{3}$ (B) $3\sqrt{5}$ (C) $4\sqrt{5}$ (D) $\frac{10\sqrt{5}}{3}$

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Q39. MCQ (Single Correct Answer)

The 20th term from the end of the progression $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \ldots, -129\frac{1}{4}$ is :

(A) -115 (B) -100 (C) -110 (D) -118

Open solution

Q40. MCQ (Single Correct Answer)

Let $f : \mathbf{R} - \left\{\frac{-1}{2}\right\} \to \mathbf{R}$ and $g : \mathbf{R} - \left\{\frac{-5}{2}\right\} \to \mathbf{R}$ be defined as $f(x) = \frac{2x+3}{2x+1}$ and $g(x) = \frac{|x|+1}{2x+5}$. Then, the domain of the function fog is :

(A) $\mathbf{R} - \left\{-\frac{7}{4}\right\}$ (B) $\mathbf{R}$

(C) $\mathbf{R} - \left\{-\frac{5}{2}, -\frac{7}{4}\right\}$ (D) $\mathbf{R} - \left\{-\frac{5}{2}\right\}$

Open solution

Q41. MCQ (Single Correct Answer)

If $\lim\limits_{x\to 0} \frac{3+\alpha \sin x+\beta \cos x+\log_e(1-x)}{3\tan^2 x} = \frac{1}{3}$, then $2\alpha - \beta$ is equal to :

(A) 2 (B) 1 (C) 5 (D) 7

Open solution

Q42. MCQ (Single Correct Answer)

If $y = y(x)$ is the solution curve of the differential equation $\left(x^2-4\right)\mathrm{d}y - \left(y^2-3y\right)\mathrm{d}x = 0, x > 2, y(4) = \frac{3}{2}$ and the slope of the curve is never zero, then the value of $y(10)$ equals :

(A) $\frac{3}{1+(8)^{1/4}}$ (B) $\frac{3}{1-(8)^{1/4}}$ (C) $\frac{3}{1-2\sqrt{2}}$ (D) $\frac{3}{1+2\sqrt{2}}$

Open solution

Q43. MCQ (Single Correct Answer)

If $2\tan^2\theta - 5\sec\theta = 1$ has exactly 7 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, for the least value of $n \in \mathbf{N}$, then $\sum\limits_{k=1}^{n} \frac{k}{2^k}$ is equal to:

(A) $\frac{1}{2^{14}}\left(2^{15}-15\right)$ (B) $1 - \frac{15}{2^{13}}$

(C) $\frac{1}{2^{15}}\left(2^{14}-14\right)$ (D) $\frac{1}{2^{13}}\left(2^{14}-15\right)$

Open solution

Q44. MCQ (Single Correct Answer)

Let $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$ and $f''(x) > 0$ for all $x \in (0,3)$. If g is decreasing in $(0,\alpha)$ and increasing in $(\alpha, 3)$, then 8α is :

(A) 0 (B) 24 (C) 18 (D) 20

Open solution

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Q45. MCQ (Single Correct Answer)

Let R be the interior region between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin. The set of all values of a, for which the points $\left(a^2, a+1\right)$ lie in R, is :

(A) $(-3,0) \cup \left(\frac{2}{3}, 1\right)$

(B) $(-3,0) \cup \left(\frac{1}{3}, 1\right)$

(C) $(-3,-1) \cup \left(\frac{1}{3}, 1\right)$

(D) $(-3,-1) \cup \left(-\frac{1}{3}, 1\right)$

Open solution

Q46. MCQ (Single Correct Answer)

Let $\alpha = \frac{(4!)!}{(4!)^{3!}}$ and $\beta = \frac{(5!)!}{(5!)^{4!}}$. Then :

(A) $\alpha \in \mathbf{N}$ and $\beta \in \mathbf{N}$

(B) $\alpha \in \mathbf{N}$ and $\beta \notin \mathbf{N}$

(C) $\alpha \notin \mathbf{N}$ and $\beta \in \mathbf{N}$

(D) $\alpha \notin \mathbf{N}$ and $\beta \notin \mathbf{N}$

Open solution

Q47. MCQ (Single Correct Answer)

The integral $\int \frac{\left(x^8 - x^2\right)\mathrm{d}x}{\left(x^{12} + 3x^6 + 1\right)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$ is equal to :

(A) $\log_{\mathrm{e}}\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/3} + \mathrm{C}$

(B) $\log_{\mathrm{e}}\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right) + \mathrm{C}$

(C) $\log_{\mathrm{e}}\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/2} + \mathrm{C}$

(D) $\log_{\mathrm{e}}\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{3} + \mathrm{C}$

Open solution

Q48. MCQ (Single Correct Answer)

The values of α, for which $\begin{vmatrix} 1 & \frac{3}{2} & \alpha + \frac{3}{2} \\ 1 & \frac{1}{3} & \alpha + \frac{1}{3} \\ 2\alpha + 3 & 3\alpha + 1 & 0 \end{vmatrix} = 0$, lie in the interval

(A) $(-2,1)$

(B) $\left(-\frac{3}{2}, \frac{3}{2}\right)$

(C) $(-3,0)$

(D) $(0,3)$

Open solution

Q49. MCQ (Single Correct Answer)

For $0 < \mathrm{a} < 1$, the value of the integral $\int_0^{\pi} \frac{\mathrm{d}x}{1 - 2\mathrm{a}\cos x + \mathrm{a}^2}$ is :

(A) $\frac{\pi^2}{\pi + a^2}$

(B) $\frac{\pi^2}{\pi - a^2}$

(C) $\frac{\pi}{1 - \mathrm{a}^2}$

(D) $\frac{\pi}{1 + \mathrm{a}^2}$

Open solution

Q50. MCQ (Single Correct Answer)

The position vectors of the vertices A, B and C of a triangle are $2\hat{i} - 3\hat{j} + 3\hat{k}, 2\hat{i} + 2\hat{j} + 3\hat{k}$ and $-\hat{i} + \hat{j} + 3\hat{k}$ respectively. Let l denote the length of the angle bisector AD of $\angle\mathrm{BAC}$ where D is on the line segment BC, then $2l^2$ equals :

(A) 45

(B) 50

(C) 42

(D) 49

Open solution

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Q51. Numerical

The mean and standard deviation of 15 observations were found to be 12 and 3 respectively. On rechecking it was found that an observation was read as 10 in place of 12 . If μ and σ^2 denote the mean and variance of the correct observations respectively, then $15\left(\mu+\mu^2+\sigma^2\right)$ is equal to _____.

Open solution

Q52. Numerical

The coefficient of x^{2012} in the expansion of $(1-x)^{2008}\left(1+x+x^2\right)^{2007}$ is equal to _____.

Open solution

Q53. Numerical

The lines $\frac{x-2}{2}=\frac{y}{-2}=\frac{z-7}{16}$ and $\frac{x+3}{4}=\frac{y+2}{3}=\frac{z+2}{1}$ intersect at the point P. If the distance of P from the line $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-1}{1}$ is l, then $14l^2$ is equal to _____.

Open solution

Q54. Numerical

Let $f(x)=\int_0^x g(t)\log_e\left(\frac{1-t}{1+t}\right)dt$, where g is a continuous odd function. If $\int_{-\pi/2}^{\pi/2}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)dx=\left(\frac{\pi}{\alpha}\right)^2-\alpha$, then α is equal to _____.

Open solution

Q55. Numerical

If the area of the region $\left\{(x,y):0\le y\le\min\left\{2x,6x-x^2\right\}\right\}$ is A, then 12 A is equal to _____.

Open solution

Q56. Numerical

If the sum of squares of all real values of α, for which the lines $2x-y+3=0, 6x+3y+1=0$ and $\alpha x+2y-2=0$ do not form a triangle is p, then the greatest integer less than or equal to p is _____.

Open solution

Q57. Numerical

Let A be a 2×2 real matrix and I be the identity matrix of order 2. If the roots of the equation $|A-xI|=0$ be -1 and 3, then the sum of the diagonal elements of the matrix A^2 is

Open solution

Q58. Numerical

Consider a circle $(x-\alpha)^2+(y-\beta)^2=50$, where $\alpha,\beta>0$. If the circle touches the line $y+x=0$ at the point P, whose distance from the origin is $4\sqrt{2}$, then $(\alpha+\beta)^2$ is equal to _____.

Open solution

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Q59. Numerical

Let the complex numbers α and $\frac{1}{\bar{\alpha}}$ lie on the circles $|z-z_0|^2=4$ and $|z-z_0|^2=16$ respectively, where $z_0=1+i$. Then, the value of $100|\alpha|^2$ is _____.

Open solution

Q60. Numerical

If the solution curve, of the differential equation $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ passing through the point $(2,1)$ is $\tan^{-1}\left(\frac{y-1}{x-1}\right)-\frac{1}{\beta}\log_e\left(\alpha+\left(\frac{y-1}{x-1}\right)^2\right)=\log_e|x-1|$, then $5\beta+\alpha$ is equal to _____.

Open solution

Physics 30 questions

Q61. MCQ (Single Correct Answer)

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In Vernier calliper if positive zero error exists, then while taking measurements, the reading taken will be more than the actual reading.

Reason (R) : The zero error in Vernier Calliper might have happened due to manufacturing defect or due to rough handling.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Open solution

Q62. MCQ (Single Correct Answer)

A ball suspended by a thread swings in a vertical plane so that its magnitude of acceleration in the extreme position and lowest position are equal. The angle (θ) of thread deflection in the extreme position will be :

(A) $\tan^{-1}\left(\frac{1}{2}\right)$

(B) $2\tan^{-1}\left(\frac{1}{2}\right)$

(C) $2\tan^{-1}\left(\frac{1}{\sqrt{5}}\right)$

(D) $\tan^{-1}(\sqrt{2})$

Open solution

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Q63. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I) : The limiting force of static friction depends on the area of contact and independent of materials.

Statement (II) : The limiting force of kinetic friction is independent of the area of contact and depends on materials.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

Q64. MCQ (Single Correct Answer)

Open solution

Primary side of a transformer is connected to $230\ \mathrm{V}, 50\ \mathrm{Hz}$ supply. Turns ratio of primary to secondary winding is $10 : 1$. Load resistance connected to secondary side is 46Ω. The power consumed in it is :

(A) 11.5 W (B) 12.5 W (C) 10.0 W (D) 12.0 W

Q65. MCQ (Single Correct Answer)

Open solution

A heavy iron bar of weight $12\ \mathrm{kg}$ is having its one end on the ground and the other on the shoulder of a man. The rod makes an angle 60° with the horizontal, the weight experienced by the man is :

(A) $3\ \mathrm{kg}$ (B) $6\ \mathrm{kg}$ (C) $6\sqrt{3}\ \mathrm{kg}$ (D) $12\ \mathrm{kg}$

Q66. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The angular speed of the moon in its orbit about the earth is more than the angular speed of the earth in its orbit about the sun.

Reason (R) : The moon takes less time to move around the earth than the time taken by the earth to move around the sun.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(B) (A) is correct but (R) is not correct

(C) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(D) (A) is not correct but (R) is correct

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Q67. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The property of body, by virtue of which it tends to regain its original shape when the external force is removed, is Elasticity.

Reason (R) : The restoring force depends upon the bonded inter atomic and inter molecular force of solid.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is false but (R) is true

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) Both (A) and (R) are true and (R) is the correct explanation of (A)

(D) (A) is true but (R) is false

Q68. MCQ (Single Correct Answer)

Open solution

The atomic mass of ${}_6C^{12}$ is $12.000000\ \mathrm{u}$ and that of ${}_6C^{13}$ is $13.003354\ \mathrm{u}$. The required energy to remove a neutron from ${}_6C^{13}$, if mass of neutron is $1.008665\ \mathrm{u}$, will be :

(A) 62.5 MeV **(B)** 6.25 MeV **(C)** 4.95 MeV **(D)** 49.5 MeV

Q69. MCQ (Single Correct Answer)

Open solution

The threshold frequency of a metal with work function $6.63\ \mathrm{eV}$ is :

(A) $16 \times 10^{15}\ \mathrm{Hz}$ **(B)** $16 \times 10^{12}\ \mathrm{Hz}$ **(C)** $1.6 \times 10^{15}\ \mathrm{Hz}$ **(D)** $1.6 \times 10^{12}\ \mathrm{Hz}$

Q70. MCQ (Single Correct Answer)

Open solution

During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio of $\frac{\mathrm{Cp}}{\mathrm{Cv}}$ for the gas is :

(A) $\frac{7}{5}$ **(B)** $\frac{3}{2}$ **(C)** $\frac{9}{7}$ **(D)** $\frac{5}{3}$

Q71. MCQ (Single Correct Answer)

Open solution

A bullet is fired into a fixed target loses one third of its velocity after travelling $4\ \mathrm{cm}$. It penetrates further $\mathrm{D} \times 10^{-3}\ \mathrm{m}$ before coming to rest. The value of D is :

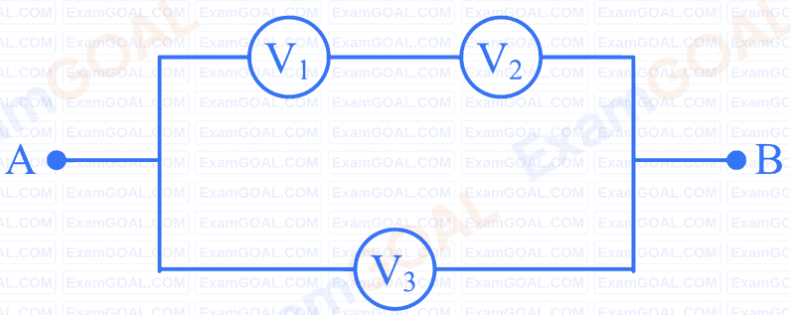
(A) 23 **(B)** 32 **(C)** 42 **(D)** 52

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Q72. MCQ (Single Correct Answer)

Three voltmeters, all having different internal resistances are joined as shown in figure. When some potential difference is applied across A and B, their readings are V_1, V_2 and V_3. Choose the correct option.

Open solution



(A) $V_1 = V_2$ (B) $V_1 \neq V_3 - V_2$ (C) $V_1 + V_2 = V_3$ (D) $V_1 + V_2 > V_3$

Q73. MCQ (Single Correct Answer)

The equation of state of a real gas is given by $\left(\mathrm{P} + \frac{\mathrm{a}}{\mathrm{V}^2}\right)(\mathrm{V} - \mathrm{b}) = \mathrm{RT}$, where P, V and T are pressure, volume and temperature respectively and R is the universal gas constant. The dimensions of $\frac{\mathrm{a}}{\mathrm{b}^2}$ is similar to that of :

Open solution

(A) P (B) RT (C) PV (D) R

Q74. MCQ (Single Correct Answer)

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Work done by electric field on moving a positive charge on an equipotential surface is always zero.

Reason (R) : Electric lines of forces are always perpendicular to equipotential surfaces.

In the light of the above statements, choose the most appropriate answer from the options given below :

Open solution

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) (A) is correct but (R) is not correct

(C) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(D) (A) is not correct but (R) is correct

Q75. MCQ (Single Correct Answer)

When a polaroid sheet is rotated between two crossed polaroids then the transmitted light intensity will be maximum for a rotation of :

Open solution

(A) 90° (B) 30° (C) 45° (D) 60°

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Q76. MCQ (Single Correct Answer)

An object is placed in a medium of refractive index 3 . An electromagnetic wave of intensity $6 \times 10^8\ \mathrm{W/m^2}$ falls normally on the object and it is absorbed completely. The radiation pressure on the object would be (speed of light in free space $= 3 \times 10^8\ \mathrm{m/s}$) :

Open solution

(A) $6\ \mathrm{Nm^{-2}}$ (B) $36\ \mathrm{Nm^{-2}}$ (C) $18\ \mathrm{Nm^{-2}}$ (D) $2\ \mathrm{Nm^{-2}}$

Q77. MCQ (Single Correct Answer)

The total kinetic energy of 1 mole of oxygen at $27^\circ\mathrm{C}$ is : [Use universal gas constant $(R) = 8.31\ \mathrm{J/}$ mole K]

Open solution

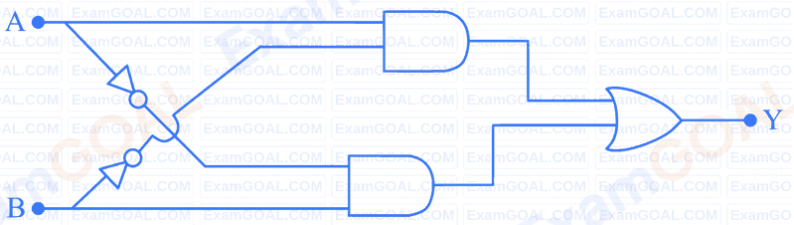
(A) 6232.5 J (B) 5670.5 J (C) 6845.5 J (D) 5942.0 J

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Q78. MCQ (Single Correct Answer)

Open solution

The truth table of the given circuit diagram is :



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(C)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(D)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Q79. MCQ (Single Correct Answer)

Open solution

A current of $200\mu A$ deflects the coil of a moving coil galvanometer through 60°. The current to cause deflection through $\frac{\pi}{10}$ radian is :

(A) 120 μA (B) 180 μA (C) 30 μA (D) 60 μA

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Q80. MCQ (Single Correct Answer)

Wheatstone bridge principle is used to measure the specific resistance (S_1) of given wire, having length L, radius r. If X is the resistance of wire, then specific resistance is ; $S_1 = X\left(\frac{\pi r^2}{L}\right)$. If the length of the wire gets doubled then the value of specific resistance will be :

(A) $\frac{S_1}{4}$ (B) $2\,S_1$ (C) $\frac{S_1}{2}$ (D) S_1

Open solution

Q81. Numerical

A closed organ pipe $150\ \text{cm}$ long gives 7 beats per second with an open organ pipe of length $350\ \text{cm}$, both vibrating in fundamental mode. The velocity of sound is _____ m/s.

Open solution

Q82. Numerical

A series LCR circuit with $\text{L} = \frac{100}{\pi}\text{mH}, \text{C} = \frac{10^{-3}}{\pi}\text{F}$ and $\text{R} = 10\Omega$, is connected across an ac source of $220\ \text{V}, 50\ \text{Hz}$ supply. The power factor of the circuit would be _____.

Open solution

Q83. Numerical

Two charges of $-4\mu\text{C}$ and $+4\mu\text{C}$ are placed at the points $\text{A}(1,0,4)\text{m}$ and $\text{B}(2,-1,5)\text{m}$ located in an electric field $\overrightarrow{\text{E}} = 0.20\hat{i}\ \text{V/cm}$. The magnitude of the torque acting on the dipole is $8\sqrt{\alpha} \times 10^{-5}\text{Nm}$, where $\alpha =$ _____.

Open solution

Q84. Numerical

A ring and a solid sphere roll down the same inclined plane without slipping. They start from rest. The radii of both bodies are identical and the ratio of their kinetic energies is $\frac{7}{x}$, where x is _____.

Open solution

Q85. Numerical

A body falling under gravity covers two points A and B separated by $80\ \text{m}$ in $2\ \text{s}$. The distance of upper point A from the starting point is _____ m (use $\text{g} = 10\ \text{ms}^{-2}$).

Open solution

Q86. Numerical

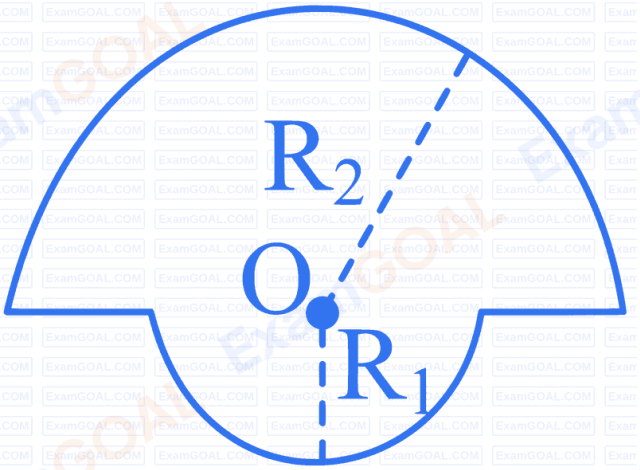
The electric potential at the surface of an atomic nucleus $(z = 50)$ of radius $9 \times 10^{-13}\ \text{cm}$ is _____ $\times 10^6\ \text{V}$.

Open solution

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Q87. Numerical

The magnetic field at the centre of a wire loop formed by two semicircular wires of radii $R_1 = 2\pi\text{m}$ and $R_2 = 4\pi\text{m}$, carrying current $\text{I} = 4\ \text{A}$ as per figure given below is $\alpha \times 10^{-7}\ \text{T}$. The value of α is _____. (Centre O is common for all segments)



Open solution

Q88. Numerical

A parallel beam of monochromatic light of wavelength 5000 $\overset{o}{A}$ is incident normally on a single narrow slit of width $0.001\ \text{mm}$. The light is focused by convex lens on screen, placed on its focal plane. The first minima will be formed for the angle of diffraction of _____ (degree).

Open solution

Q89. Numerical

The reading of pressure metre attached with a closed pipe is $4.5 \times 10^4\ \text{N/m}^2$. On opening the valve, water starts flowing and the reading of pressure metre falls to $2.0 \times 10^4\ \text{N/m}^2$. The velocity of water is found to be $\sqrt{V}\ \text{m/s}$. The value of V is _____.

Open solution

Q90. Numerical

If Rydberg's constant is R, the longest wavelength of radiation in Paschen series will be $\frac{\alpha}{7R}$, where $\alpha =$ _____.

Open solution

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Answer Key

Chemistry

1. **D**	2. **B**	3. **D**	4. **B**	5. **D**
6. **B**	7. **D**	8. **A**	9. **B**	10. **C**
11. **C**	12. **C**	13. **C**	14. **C**	15. **B**
16. **C**	17. **C**	18. **D**	19. **B**	20. **A**
21. **8**	22. **0**	23. **5**	24. **10**	25. **18**
26. **4**	27. **135**	28. **7**	29. **2**	30. **37**

Mathematics

31. **D**	32. **D**	33. **A**	34. **C**	35. **C**
36. **B**	37. **B**	38. **D**	39. **A**	40. **D**
41. **C**	42. **A**	43. **D**	44. **C**	45. **B**
46. **A**	47. **A**	48. **C**	49. **C**	50. **A**
51. **2521**	52. **0**	53. **108**	54. **2**	55. **304**
56. **32**	57. **10**	58. **100**	59. **20**	60. **11**

Physics

61. **B**	62. **B**	63. **A**	64. **A**	65. **A**
66. **C**	67. **C**	68. **C**	69. **C**	70. **B**
71. **B**	72. **C**	73. **A**	74. **A**	75. **C**
76. **A**	77. **A**	78. **A**	79. **D**	80. **D**
81. **294**	82. **1**	83. **2**	84. **7**	85. **45**
86. **8**	87. **3**	88. **30**	89. **50**	90. **144**

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